

# Dr.-Ing. Thomas Pöllabauer-Berkei

Technical Lead | Applied AI & Computer Vision

## CONTACT

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## EDUCATION

**Dr.-Ing., Computer Science**  
Mar 2023 – Jan 2025  
TU Darmstadt, Interactive Graphics Systems  
| Darmstadt, Germany  
Thesis: *Advancing ML Algorithms for Object Localization in Data-Limited Scenarios*. Awarded *magna cum laude*.

**M.Sc., Visual Computing**  
Oct 2017 – Feb 2020  
TU Darmstadt | Darmstadt, Germany

**B.Sc., Computer Science**  
Oct 2013 – Sept 2017  
TU Berlin | Berlin, Germany

**Abitur, Electronics & Technical Computer Science**  
Sept 2007 – Jun 2012  
Höhere Technische Lehranstalt Steyr | Steyr, Austria

## SKILLS

**Core Competencies**  
Machine Learning | Computer Vision | MLOps  
| Kubernetes | OpenShift | Cloud-Native AI  
| Technical Leadership | Industrial AI  
| Synthetic Data | CI/CD | Software Architecture  
| Distributed Systems

**Specialization**  
2D detection; 3- and 6-DoF object pose estimation; classification; semantic segmentation; 3D scene reconstruction; zero- and few-shot methods; generative models (Autoencoders, GANs, Diffusion); ML under limited data or compute.

**Technical**  
Python (PyTorch, TensorFlow, Keras, NumPy, Scikit-learn, Pandas); C/C++; Java; SQL; HTML/CSS; Git; Linux; Bash/Zsh; Docker; SLURM; GitOps;  $\LaTeX$ .

**Leadership**  
Project management; research design; scientific and technical writing; process docu-

## PROFILE

Technical Lead with a track record of bridging cutting-edge AI research and industrial software engineering. Experienced in leading scalable AI systems from research prototype to production, with a focus on machine learning, computer vision, and cloud-native architecture. Combines deep technical expertise with stakeholder communication and cross-functional coordination. A computer vision and machine learning enthusiast with a longstanding passion for computer science, nurtured since high school; over six years at the intersection of research and industry across automotive, engineering, and aerospace clients.

## EMPLOYMENT HISTORY

**Technical Lead** May 2025 – present  
PROFI Engineering Systems AG | Remote

- Owns technical strategy, architecture, and delivery of AI-driven software products, focused on scalable ML systems and industrial analytics – including PROinfra, predictive maintenance for civil infrastructure such as bridges, roads, and retaining walls.
- Domain expert for Machine Learning, Computer Vision, and applied AI; leads deployment on Kubernetes and OpenShift.
- Coordinates engineering, product, and business stakeholders; pre-sales and technical consulting for customers and partners.
- Hires and mentors engineering talent.

**Research Scientist, Deep Learning & Computer Vision** Apr 2020 – Dec 2024  
Fraunhofer IGD, Virtual and Augmented Reality | Darmstadt, Germany

- State-of-the-art computer vision for object detection and localization, segmentation, and classification.
- Generative ML models (GANs, encoder-decoder, diffusion); synthetic data pipelines and domain adaptation for robust AI.
- Automated industrial quality inspection; scalable training and inference infrastructure for real-world deployment.
- Led industry collaborations in manufacturing, robotics, AR/VR, automotive, and infrastructure; supervised student research; published peer-reviewed work in CV, ML, and 3D vision.

**Student Assistant, Computer Vision** Sept 2018 – Feb 2020  
Fraunhofer IGD, Virtual and Augmented Reality | Darmstadt, Germany

- Data analysis, CI and testing, ML, and optimization for Computer Vision and Visual Computing projects.
- Built experimental software prototypes and evaluation pipelines.

**Student Assistant, Editorial Staff** Oct 2013 – Sept 2014  
Topography of Terror Foundation | Berlin, Germany

**International Memorial Service** Oct 2012 – Sept 2013  
Topography of Terror Foundation | Berlin, Germany

## SELECTED PROJECTS

**Project Lead – BAU-DNS** Mar 2023 – Dec 2024  
Sustainable modular circular building renovation | Fraunhofer-wide Research Project | Darmstadt

**Project Lead – SEC-Learn** Jan 2021 – Dec 2024  
Sensor Edge Cloud for distributed learning | Fraunhofer-wide Research Project | Darmstadt

**ICE train interior surveillance PoC** Mar 2022 – Sept 2022  
Cooperation with Deutsche Bahn AG | 3rd Place, best Industrial Project, Fraunhofer IGD

**Project Lead – DiffEd-Pose** Feb 2021 – Jan 2022  
Object pose estimation from synthetic data via differentiable rendering-driven federated learning | Cooperation with Software AG

mentation; conference and fair presentation; customer contact.

LANGUAGES

German: native  
English: professional proficiency

RESEARCH

Doctorate: Dr.-Ing., magna cum laude, TU Darmstadt  
Publications: 20+ peer-reviewed papers in Computer Vision and Machine Learning; see Google Scholar and ORCID

ACHIEVEMENTS

2. BMW-Fraunhofer Research Summit March 2023  
Awarded best pitch by jury; invited to the BMW-Fraunhofer Technologietag

Software Campus Program to promote IT talent Feb 2021 - June 2022  
Funded with 100,000€ by the Bundesministerium für Bildung und Forschung